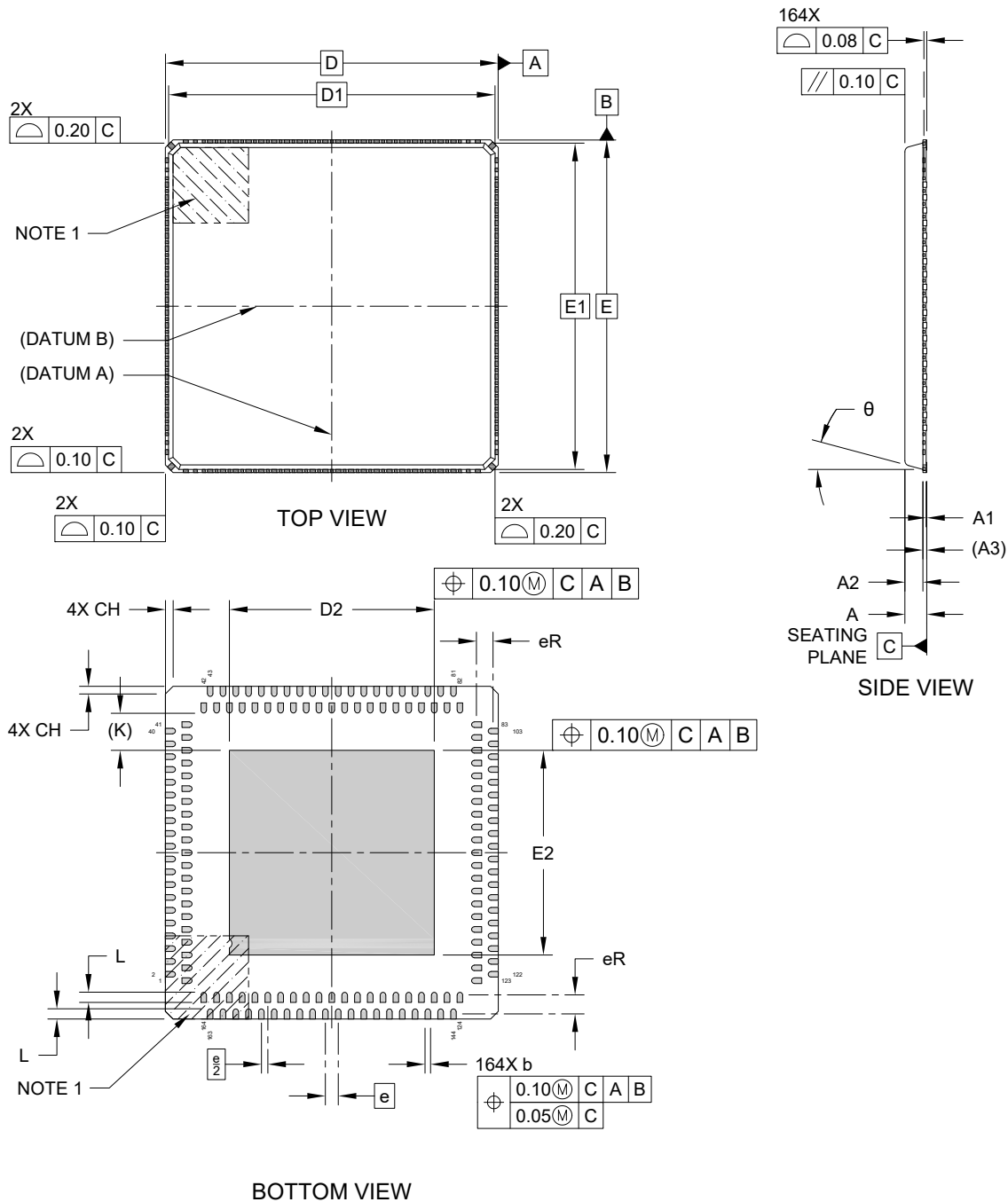


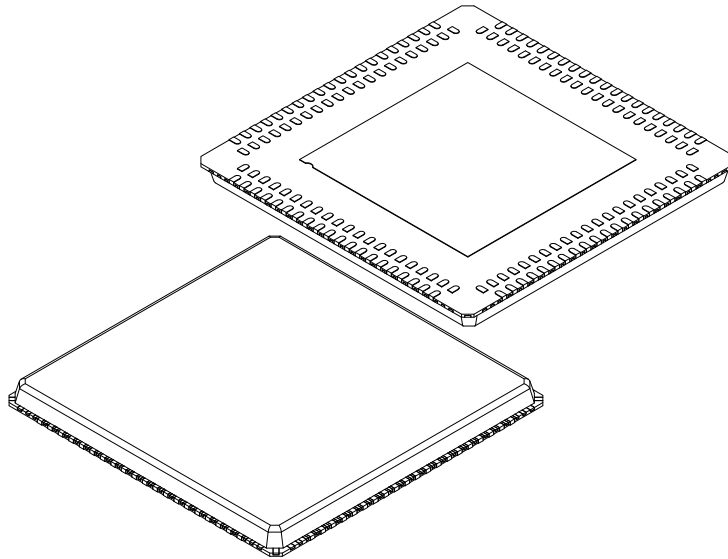
# 164-Lead Very Thin Plastic Quad Flat, No Lead Package (PMX) - 13x13x0.9 mm Body [VQFN]; Dual Row, Punch Singulated

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**164-Lead Very Thin Plastic Quad Flat, No Lead Package (PMX) - 13x13x0.9 mm Body [VQFN]; Dual Row, Punch Singulated**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



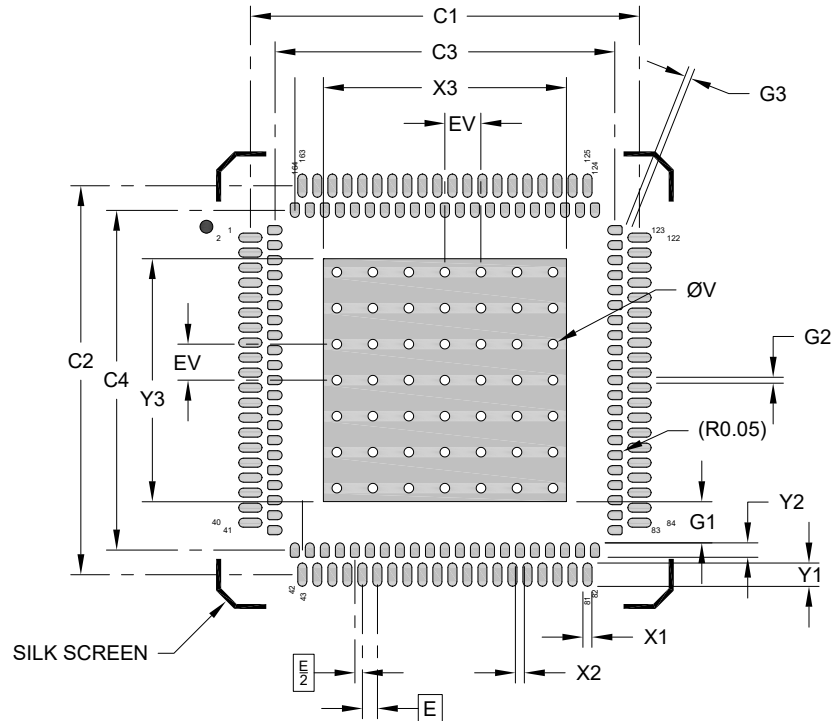
| Units                   |    | MILLIMETERS |      |      |
|-------------------------|----|-------------|------|------|
| Dimension Limits        |    | MIN         | NOM  | MAX  |
| Number of Terminals     | N  | 164         |      |      |
| Pitch                   | e  | 0.50 BSC    |      |      |
| Overall Height          | A  | –           | –    | 0.90 |
| Standoff                | A1 | 0.00        | 0.02 | 0.05 |
| Molded Package Height   | A2 | –           | 0.70 | 0.75 |
| Terminal Thickness      | A3 | 0.152 REF   |      |      |
| Overall Length          | D  | 13.00 BSC   |      |      |
| Molded Package Length   | D1 | 12.75 BSC   |      |      |
| Exposed Pad Length      | D2 | 7.90        | 8.00 | 8.10 |
| Overall Width           | E  | 13.00 BSC   |      |      |
| Molded Package Width    | E1 | 12.75 BSC   |      |      |
| Exposed Pad Width       | E2 | 7.90        | 8.00 | 8.10 |
| Terminal Row Spacing    | eR | 0.65 BSC    |      |      |
| Terminal Width          | b  | 0.18        | 0.22 | 0.30 |
| Terminal Length         | L  | 0.30        | 0.40 | 0.50 |
| Terminal-to-Exposed-Pad | K  | 1.45 REF    |      |      |
| Package Corner Chamfer  | CH | –           | –    | 0.60 |
| Mold Draft Angle        | θ  | 5°          | –    | 15°  |

**Notes:**

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is punch singulated
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

# **164-Lead Very Thin Plastic Quad Flat, No Lead Package (PMX) - 13x13x0.9 mm Body [VQFN]; Dual Row, Punch Singulated**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



**RECOMMENDED LAND PATTERN**

| Dimension                              | Units | MILLIMETERS |          |      |
|----------------------------------------|-------|-------------|----------|------|
|                                        |       | MIN         | NOM      | MAX  |
| Contact Pitch                          | E     |             | 0.50 BSC |      |
| Center Pad Width                       | X3    |             |          | 8.10 |
| Center Pad Length                      | Y3    |             |          | 8.10 |
| Outer Row Contact Pad Spacing          | C1    |             | 12.96    |      |
| Outer Row Contact Pad Spacing          | C2    |             | 12.96    |      |
| Inner Row Contact Pad Spacing          | C3    |             | 11.32    |      |
| Inner Row Contact Pad Spacing          | C4    |             | 11.32    |      |
| Outer Contact Pad Width (X80)          | X1    |             |          | 0.30 |
| Outer Contact Pad Length (X80)         | Y1    |             |          | 0.78 |
| I84er Contact Pad Width (X84)          | X2    |             |          | 0.30 |
| I84er Contact Pad Length (X84)         | Y2    |             |          | 0.48 |
| Contact Pad to Center Pad              | G1    | 1.35        |          |      |
| Contact Pad to Contact Pad             | G2    | 0.20        |          |      |
| Outer Contact Pad to Inner Contact Pad | G3    | 0.20        |          |      |
| Thermal Via Diameter                   | V     |             | 0.33     |      |
| Thermal Via Pitch                      | EV    |             | 0.20     |      |

**Notes:**

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2527 Rev D